Kaveri Precision Components Pvt. Ltd. - Hosur Plant

Incoming Material Inspection - Standard Operating Procedure

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Prepared By	K. Divya, Supplier Quality Engineer	Approved By	M. Prakash, QC Manager

1. Purpose

This procedure describes how incoming raw materials and bought-out components are inspected at the receiving area before they are released to stores and production.

2. Scope

Applies to all steel bar stock, castings, forgings and bought-out components received at the Hosur plant, including machined BH-220 bearing housings received from sub-contractors.

3. Responsibilities

- **Stores Executive:** receives material, checks quantity against the delivery challan and raises the inspection request in the QMS app.
- **QC Inspector:** carries out sampling, visual and dimensional inspection, and records results in the QMS app.
- **Shift QC Supervisor:** approves acceptance, rejection or hold of the lot.
- **Supplier Quality Engineer (SQE):** handles supplier communication for rejected or held lots.

4. Sampling Plan

Defects are classified as follows:

- **Critical:** a defect that can cause a safety risk or failure of the part, e.g. crack, wrong material grade, or a dimension out of tolerance on a characteristic marked **Critical**.
- **Major:** a defect that affects fit or function but is not critical, e.g. deep dent, damaged thread, or a dimension out of tolerance on a characteristic marked **Major**.
- **Minor:** a cosmetic defect that does not affect fit or function, e.g. light surface marks or light surface rust that can be removed by cleaning.

The number of samples depends on the lot size. Minor defects are allowed only as shown below.

Lot size (pieces)	Samples to inspect	Accept if minor defects are	Reject if minor defects are
2 - 5	All pieces	0	1 or more
6 - 90	5	0	1 or more
91 - 500	13	0	1 or more
501 - 3,200	32	1 or fewer	2 or more
Above 3,200	50	2 or fewer	3 or more

Any critical defect in any sample: the complete lot is rejected and returned to the supplier. Sorting is not allowed.

Any major defect in any sample: the complete lot is rejected. The SQE may decide to have the lot 100% sorted at the supplier's cost instead of returning it.

5. Procedure

1. Stores Executive places the received lot in the Incoming Inspection Area and raises an inspection request in the QMS app.
2. QC Inspector checks that the supplier's **Material Test Certificate (MTC)** is attached. The MTC is **mandatory**. If it is missing, the lot is put on HOLD and must not be inspected or accepted until the MTC is received.
3. Draw samples at random as per the sampling plan in Section 4.
4. Perform visual inspection for rust, dents, cracks and burrs.
5. Perform dimensional inspection using a calibrated digital caliper. Dimensions of BH-220 housings marked CMM in Section 6 must be checked on the CMM.
6. Record results in the QMS app using digital form **QF-07A**. Paper form QF-07 is no longer used.
7. Accepted material is tagged with a GREEN tag and moved to stores.
8. Rejected material is tagged with a RED tag and moved to **Quarantine Cage Q1, Stores Bay 3**. The cage is locked; the key is held by the Shift QC Supervisor.
9. The Shift QC Supervisor must inform the SQE **within 4 hours** of rejection. The SQE raises a Supplier Corrective Action Request (SCAR).

6. Inspection Dimensions - BH-220 Bearing Housing (Machined, from sub-contractor)

All values in millimetres. Dimensions marked CMM must be checked on the coordinate measuring machine.

Characteristic	Nominal	Tolerance	Method	Class
Bore diameter	52.000	+0.030 / -0.000	CMM	Critical
Outer diameter	90.000	+/- 0.050	Digital caliper	Major
Overall height	38.00	+/- 0.10	Digital caliper	Major
Mounting hole PCD	72.00	+/- 0.05	CMM	Critical
Mounting hole diameter (4x)	9.00	+0.20 / -0.00	Pin gauge	Major
Bore concentricity to OD	-	0.02 max.	CMM	Critical

7. Records

- QF-07A Incoming Inspection Report (digital, in QMS app) - retained for 5 years.
- Supplier MTCs - scanned and attached to the QF-07A record.

8. Revision History

Rev	Date	Change
01	10-Jan-2019	First issue
02	05-Aug-2021	Added visual inspection criteria
03	15-Mar-2023	Changed sample size to 5 for all lots
04	01-Jul-2026	Supersedes Rev 03. Lot-size based sampling plan; MTC made mandatory; quarantine moved to Cage Q1; digital form QF-07A; 4-hour SQE escalation; BH-220 critical dimensions added.